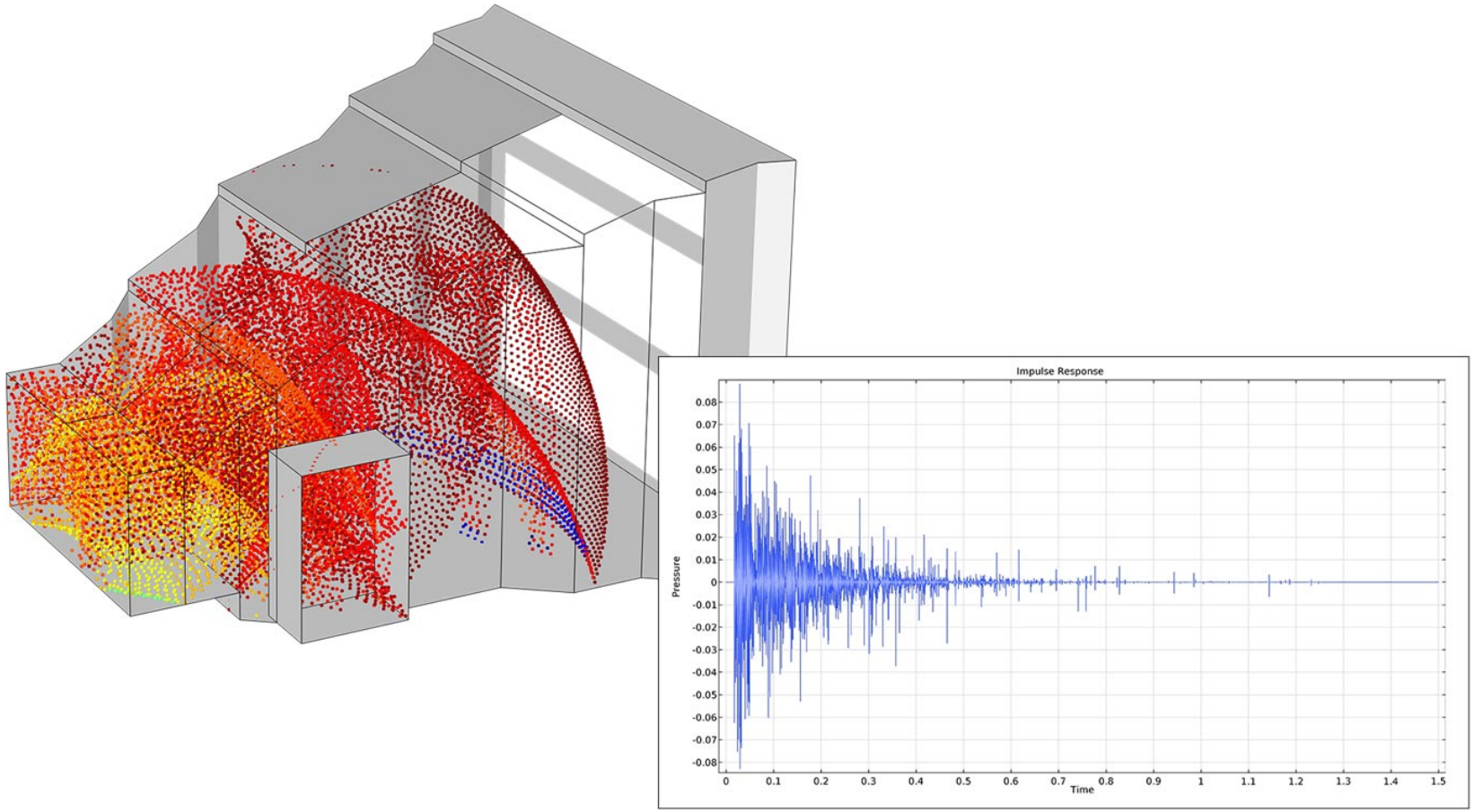
Jet marching methods for solving the eikonal equation

Samuel F. Potter
Joint work with Dr. Maria Cameron

UMD Dept. of Computer Science

December 7, 2020

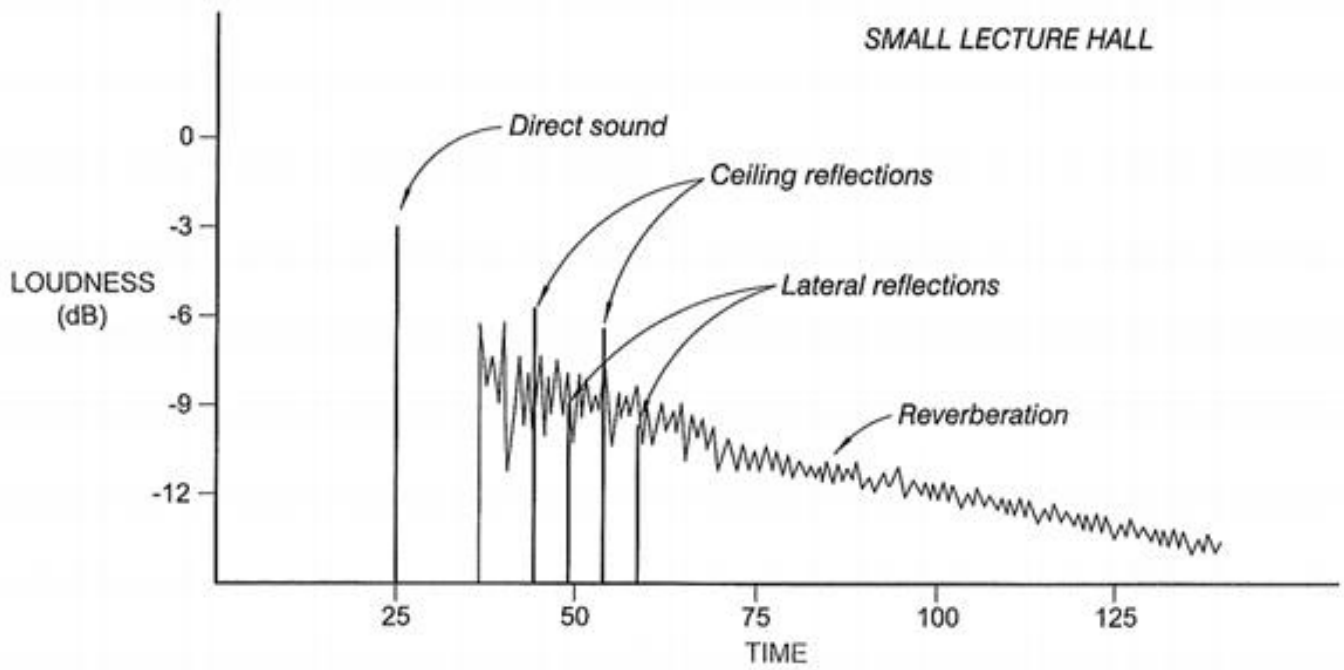
Computational room acoustics



Source: `comsol.com`

Precomputed sound propagation

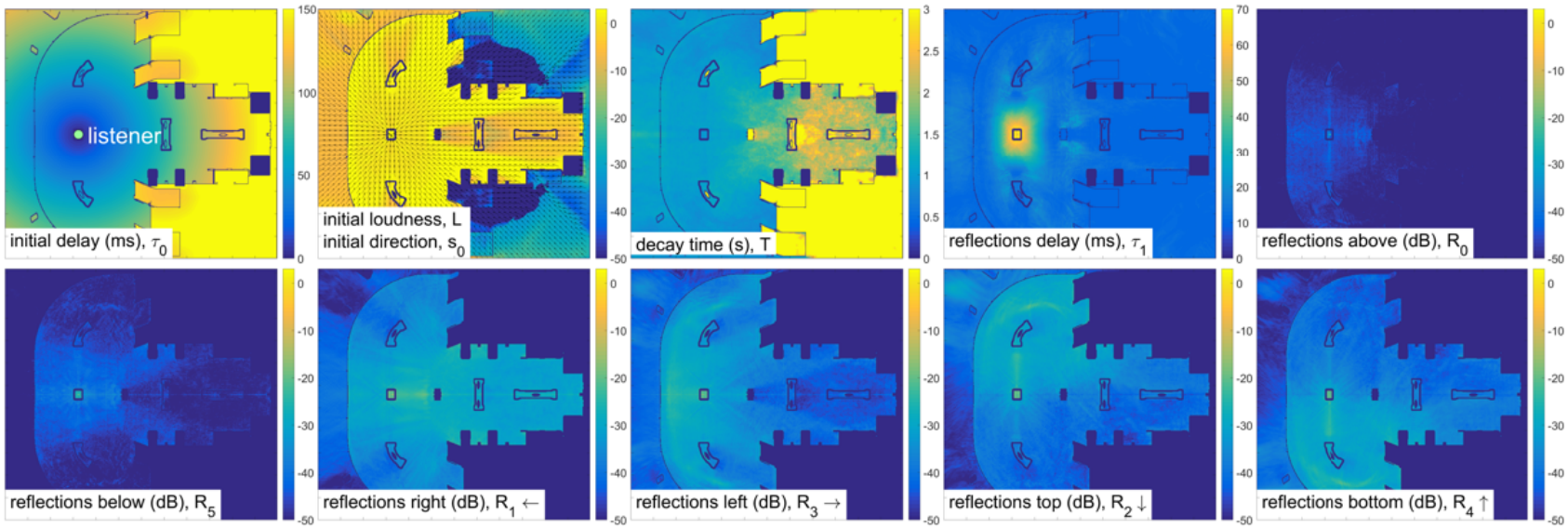
Basic task: for a pair of points in the room (a *source* and *listener*), compute or encode the room impulse response. This is the Green's function for the wave equation for a *fixed sound source location*.



Source: https://sites.google.com/site/roomacgroup4/

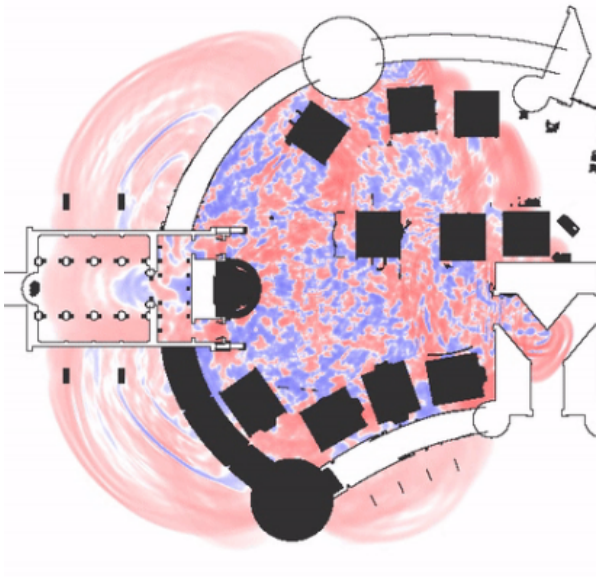
Precomputed sound propagation

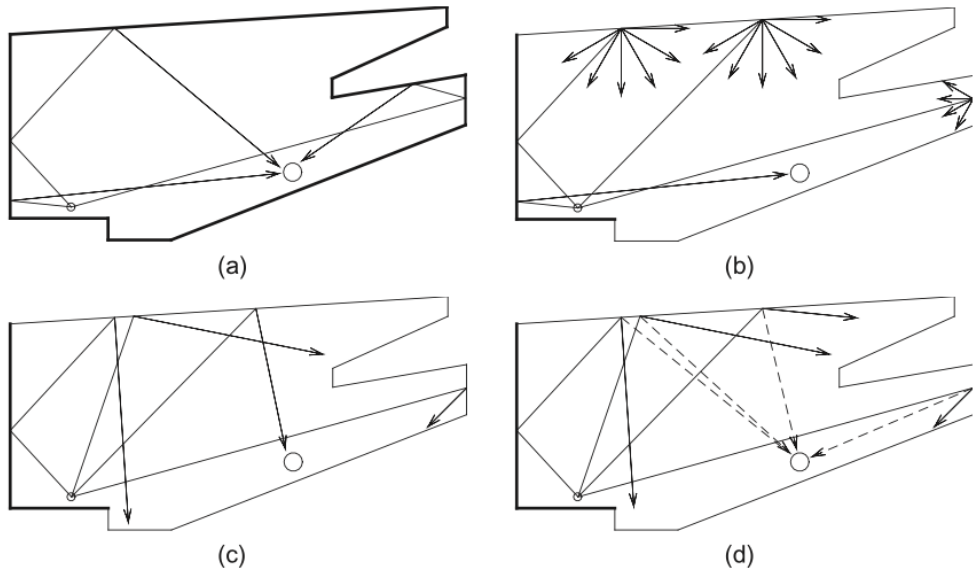
Too expensive to store the room impulse response for each source location. Instead, extract smooth fields containing **important perceptual parameters** and compress these.



Source: `https://www.microsoft.com/en-us/research/project/project-triton`

Numerical vs. geometric acoustics





Numerical acoustics: solve wave equation for bandlimited impulse input. Eulerian.

Geometric acoustics: use raytracing to assemble room impulse response. Lagrangian.

Source: `https://www.microsoft.com/en-us/research/project/project-triton`

Geometric optics from Helmholtz

For speed of sound c and frequency $\omega \gg 0$, the *Helmholtz equation* is:

$$\left(\Delta + \frac{\omega^2}{c^2}\right) U = 0$$

High-frequency geometric optics ansatz (WKB) is:

$$U(x) = \alpha(x) e^{i\omega\tau(x)} + O(\omega^{-1})$$

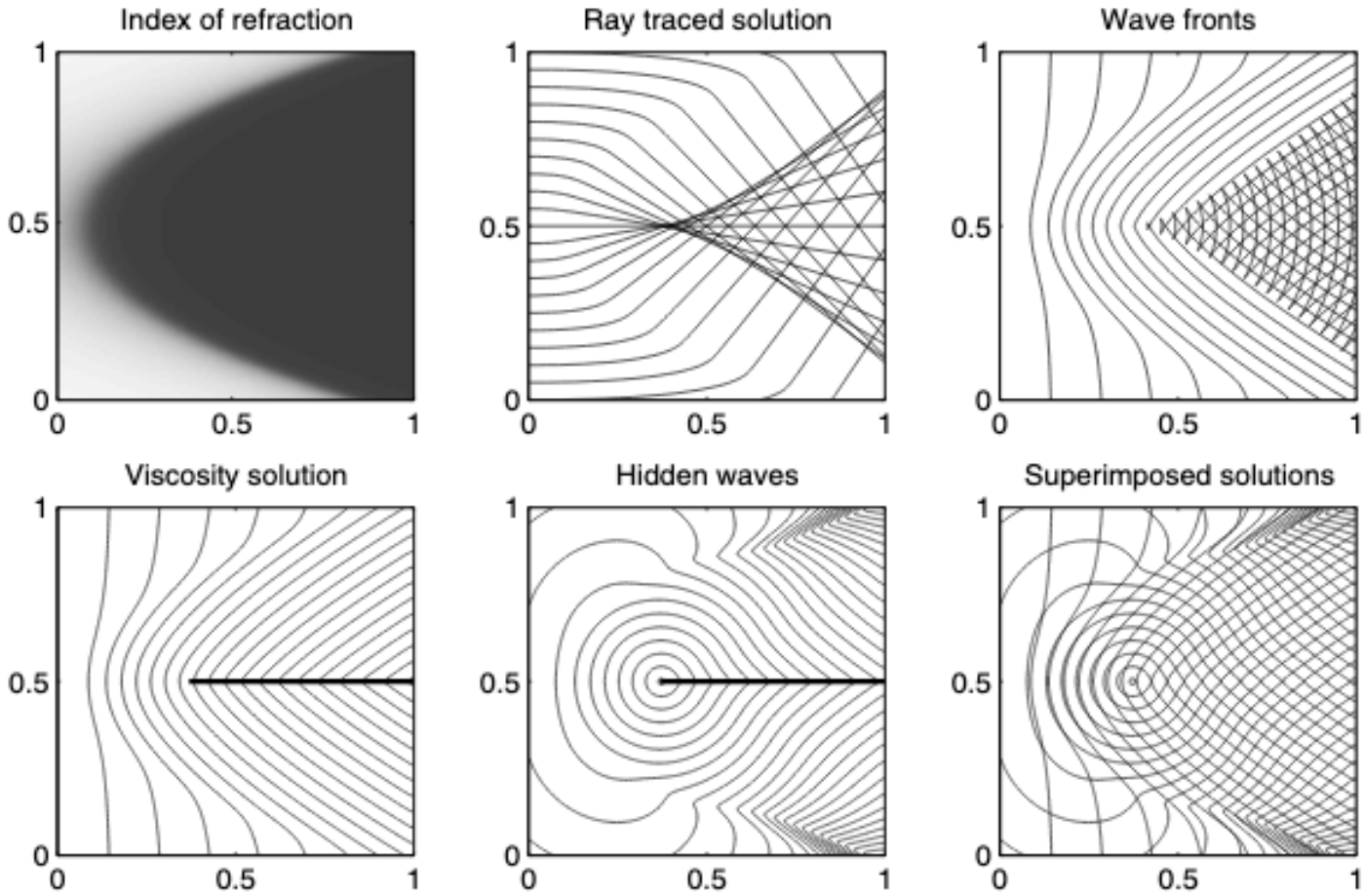
Here τ satisfies eikonal equation:

$$\|\nabla\tau\| = 1/c$$

and α is amplitude, which satisfies the *transport equation*:

$$\alpha\Delta\tau + 2\nabla\alpha^\top\nabla\tau = 0$$

“Numerical geometric acoustics”



Geometric acoustics using semi-Lagrangian numerical algorithms (“Eulerian geometric optics”)

Source: *“Computational high frequency wave propagation”*, Engquist and Runborg.

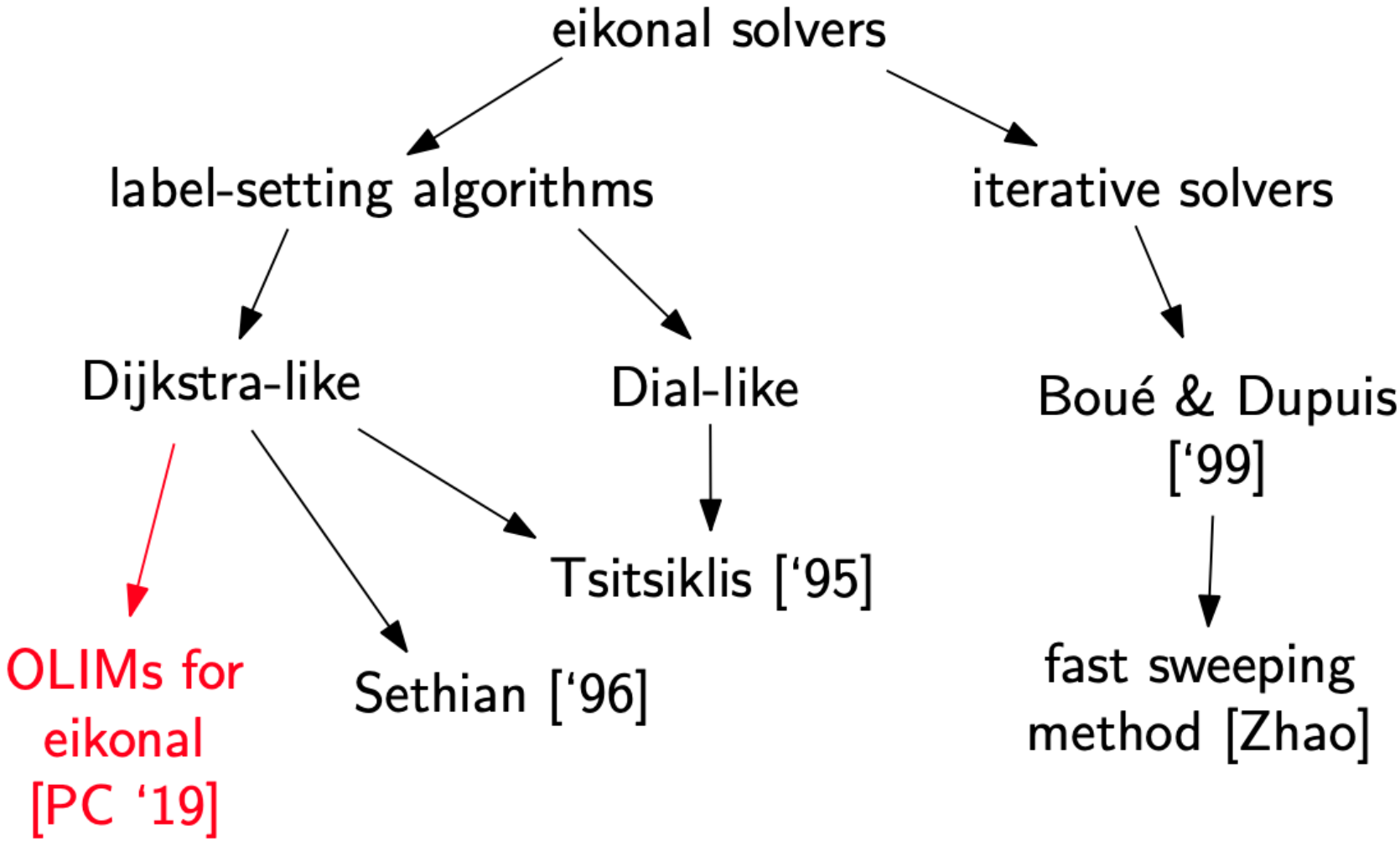
Desiderata

We need an eikonal solver that:

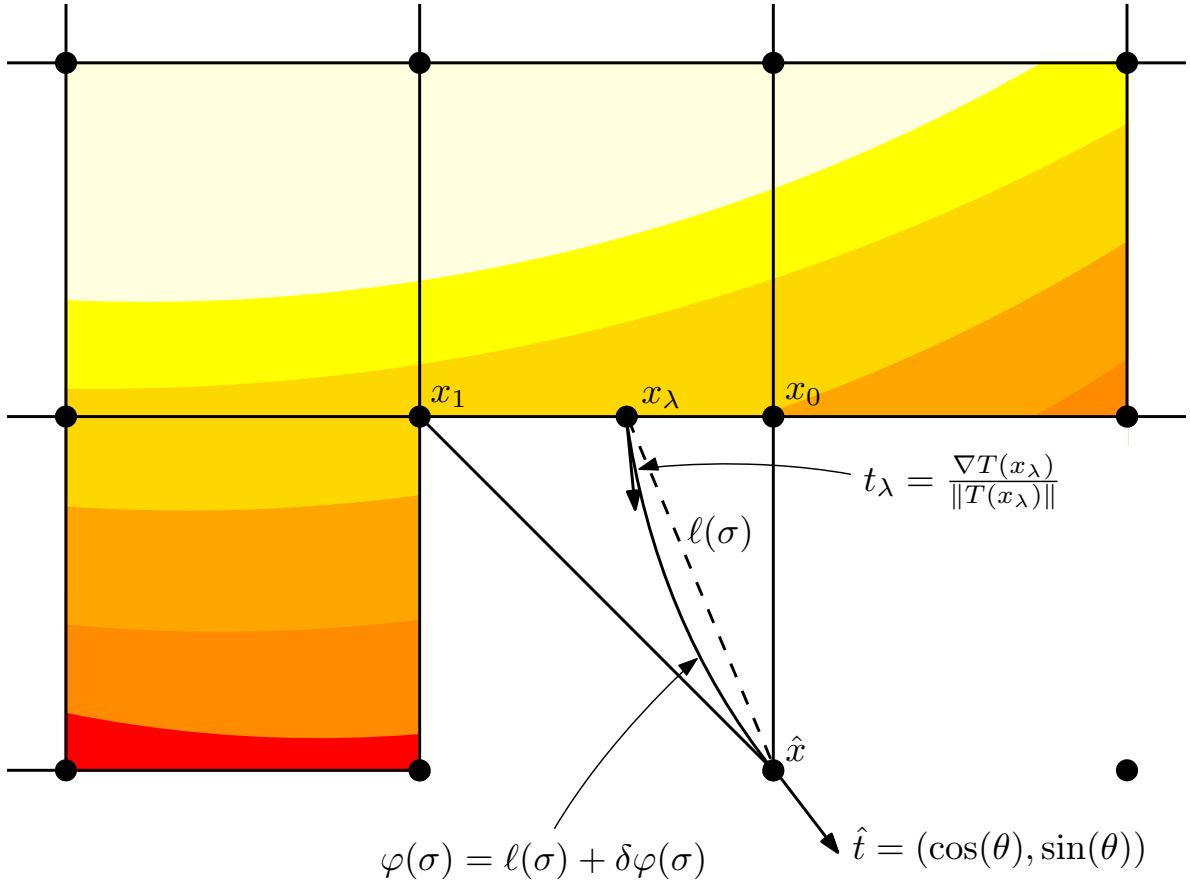
- Is at least $O(h^{3-p})$ accurate for $\partial^\alpha \tau$, $|\alpha| = p$
- Has a *compact stencil*
- Parametrizes local characteristics
- Is causal/is a direct solver
- Runs in optimal time ($O(N)$ or $O(N \log N)$ time)

Key idea: **design solver based on Hermite interpolation, semi-Lagrangian updates, and cell marching**

Prior art


eikonal solvers
label-setting algorithms
iterative solvers
Dijkstra-like
Dial-like
Boué & Dupuis ['99]
Tsitsiklis ['95]
OLIMs for eikonal [PC '19]
Sethian ['96]
fast sweeping method [Zhao]

Jet marching methods for the eikonal equation



Fermat's principle:

$$\tau(\hat{x}) = \min_{x,\psi} \left\{ \tau(x) + \int_0^1 s(\psi(\sigma)) \|\psi'(\sigma)\| d\sigma \right\}$$

Nice video here: `https://youtu.be/Ze9AeDbaDVM`

Jet marching methods for the eikonal equation

Framework:

- Build up data for polynomial approx. of τ over $[x_0, x_1]$
- Approximate local ray φ with polynomial
- Approximate action integral with high order quadrature

Hermite + Hermite + Simpson's rule yields:

$$T(\hat{x}) = \min_{\substack{\hat{t}\in\mathbb{S}^1 \\ 0\leq\lambda\leq 1}} \left\{ \mathsf{T}(\lambda) + \frac{L}{6}\left(s(\hat{x}) + 4s(\varphi(\tfrac{L}{2}))\|\varphi'(\tfrac{L}{2})\| + s(x_\lambda) \right) \right\}$$

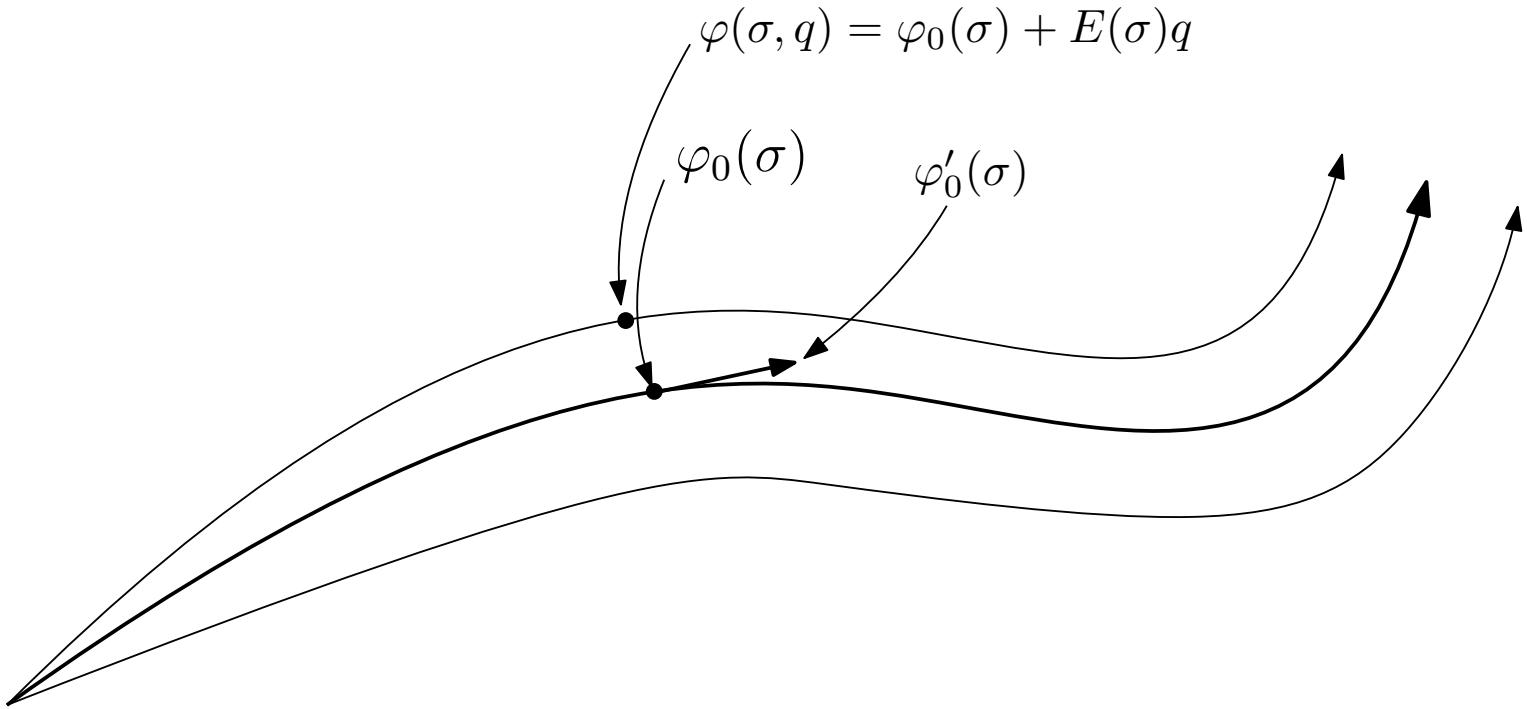
Solve w/ Newton's method using warm start.

We consider eight different variations on this idea. More details in paper.

Source: "Jet marching methods for solving the eikonal equation", Potter and Cameron, *in submission*.

Paraxial ray theory

Reminder: $U \sim \alpha e^{i\omega\tau}$. How do we march α?



Parametrize a ray by:

$$\varphi(\sigma, q) = \varphi_0(\sigma) + E(\sigma)q, \qquad E^\top \varphi_0' \equiv 0$$

Paraxial ray theory

If q is required to satisfy Euler-Lagrange equations for the eikonal equation, then it and conjugate momenta p satisfy:

$$\begin{bmatrix} dq/d\sigma \\ dp/d\sigma \end{bmatrix} = \begin{bmatrix} & c(\varphi_0(\sigma))I \\ \frac{-1}{c(\varphi_0(\sigma))} \frac{\partial^2 c}{\partial q \partial q^\top}\Big|_{q=0} & \end{bmatrix} \begin{bmatrix} q \\ p \end{bmatrix}$$

If Q and P are a linearly independent set of solutions, then:

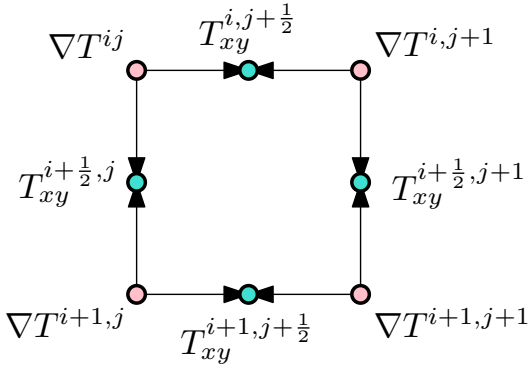
$$J \triangleq |\det Q|, \qquad A(\varphi_0(\sigma)) = \sqrt{\frac{c(\varphi_0(\sigma))}{J(\varphi_0(\sigma))}} A(\varphi_0(0))$$

Gives update:

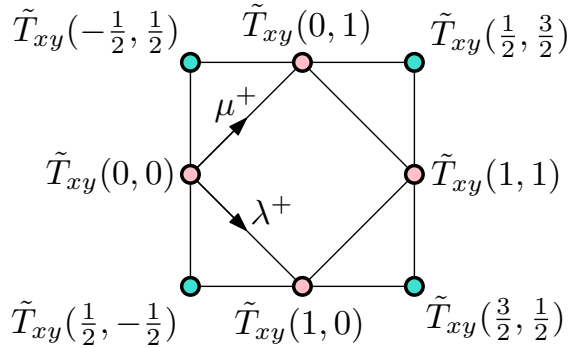
$$J(\hat{x}) = \left|1 + \epsilon \cdot \left(\Delta T(x_\lambda) - t_\lambda^\top \nabla s(x_\lambda)\right)\right| J(x_\lambda)$$

Requires us to know second partials of τ at x_λ...

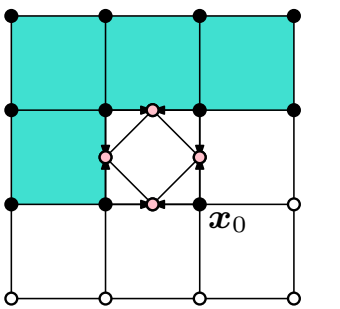
Cell marching



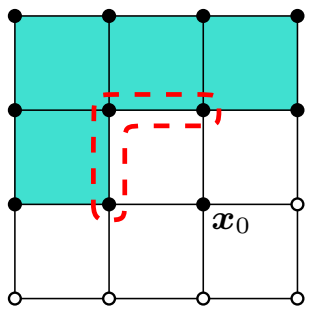
approximate T_{xy} at cell edge midpoints using central differences



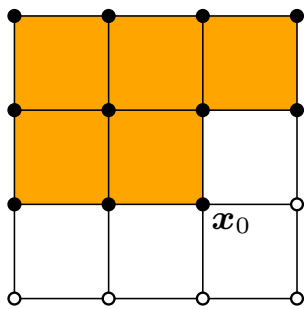
approximate T_{xy} at cell vertices using bilinear extrapolation



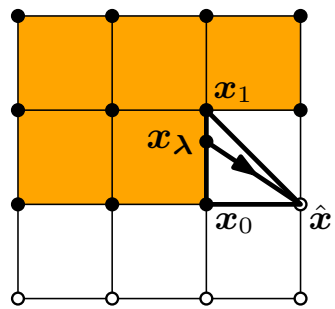
compute new T_{xy} values at corners of new `valid` cell



average T_{xy} values over adjacent `valid` cells



rebuild bicubics on affected `valid` cells



update neighboring `trial` nodes using triangle updates

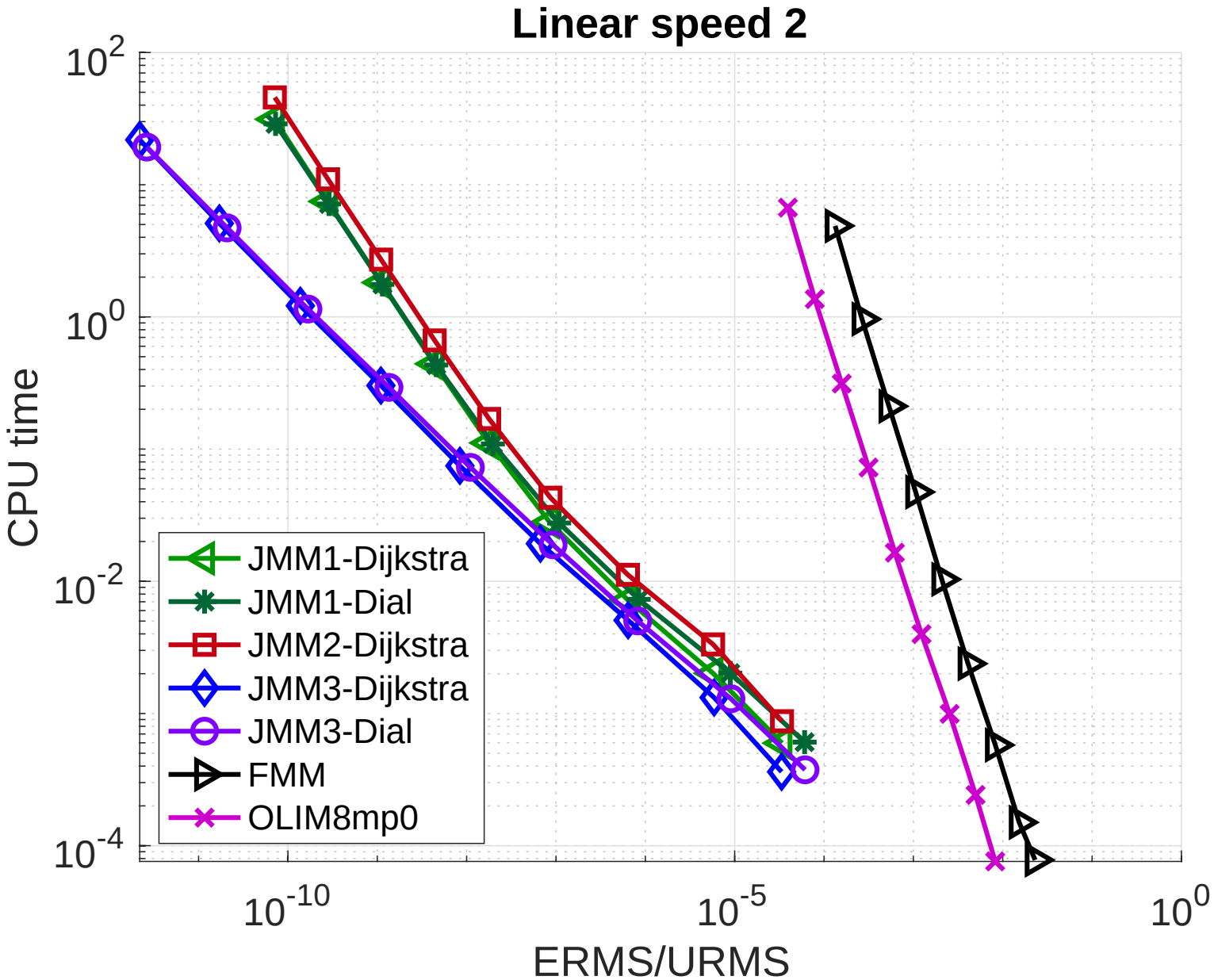
Numerical results

Point source using $s(x)$ with analytically available $\tau(x)$:

Name	$s(x)$	$\tau(x)$
Constant	1	$\|x\|$
Linear	$\left[\frac{1}{s_0} + v^\top\right]^{-1}$	$\frac{1}{\|v\|}\cosh^{-1}\left(1 + \frac{1}{2}s_0 s(x)\|v\|^2\|x\|^2\right)$
Linear #1	*	$s_0 = 1,\ v = (0.133, -0.0933)$
Linear #2	*	$s_0 = 2,\ v = (0.5, 0)$

Name	$s(x)$	$\tau(x)$
Sine	$\sqrt{\sin(x_1 + x_2)^2 + \left(x_1 + \sin(x_1 + x_2)\right)^2}$	$\frac{x_1^2}{2} + 2\sin\left(\frac{x_1 + x_2}{2}\right)^2$
Sloth	$\sqrt{s_0^2 + 2v^\top x}$	complicated

Numerical results

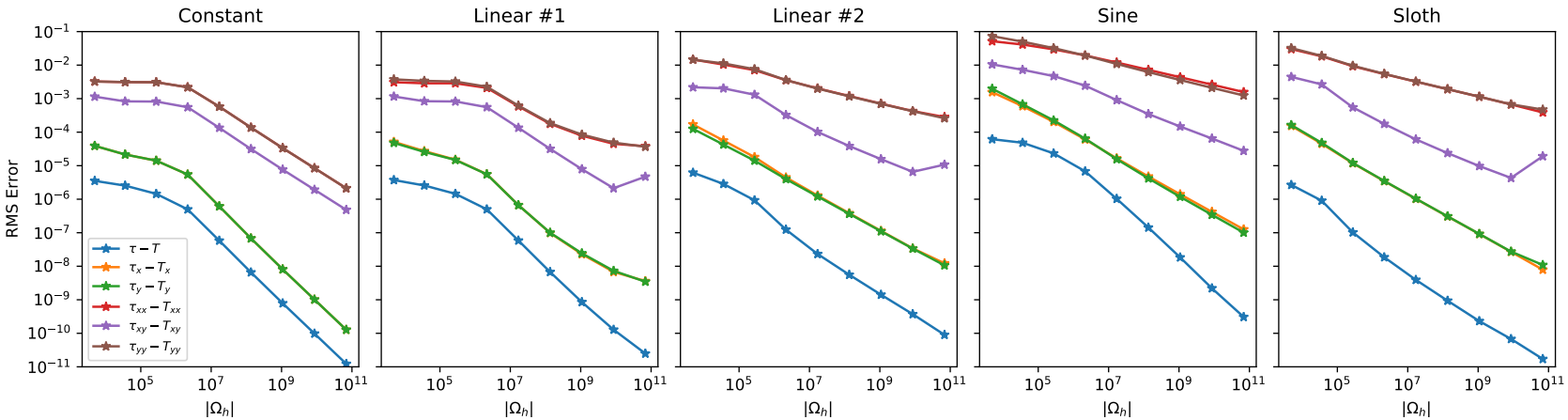


Numerical results

The order of convergence p for each combination of test problems and solvers, computed for different types of errors and fit as Ch^p.

	JMM	$E_{\max}(T)$	$E_{\mathrm{RMS}}(T)$	$E_{\max}(\nabla T)$	$E_{\mathrm{RMS}}(\nabla T)$
Constant	#1	2.87	2.87	2.28	2.72
	#2	2.87	2.87	2.28	2.72
	#3	2.87	2.87	2.28	2.72
Linear #1	#1	2.77	2.85	2.14	2.52
	#2	2.77	2.85	1.70	2.48
	#3	2.86	2.87	2.28	2.73
Linear #2	#1	2.48	2.52	1.70	1.97
	#2	2.38	2.52	1.16	1.88
	#3	3.03	3.03	2.70	3.02
Sine	#1	2.76	2.57	1.77	2.09
	#2	2.51	2.46	1.58	1.94
	#3	2.37	2.38	1.54	1.79
Sloth	#1	2.39	2.48	1.49	1.84
	#2	2.37	2.47	0.87	1.73
	#3	2.15	2.21	1.47	1.76

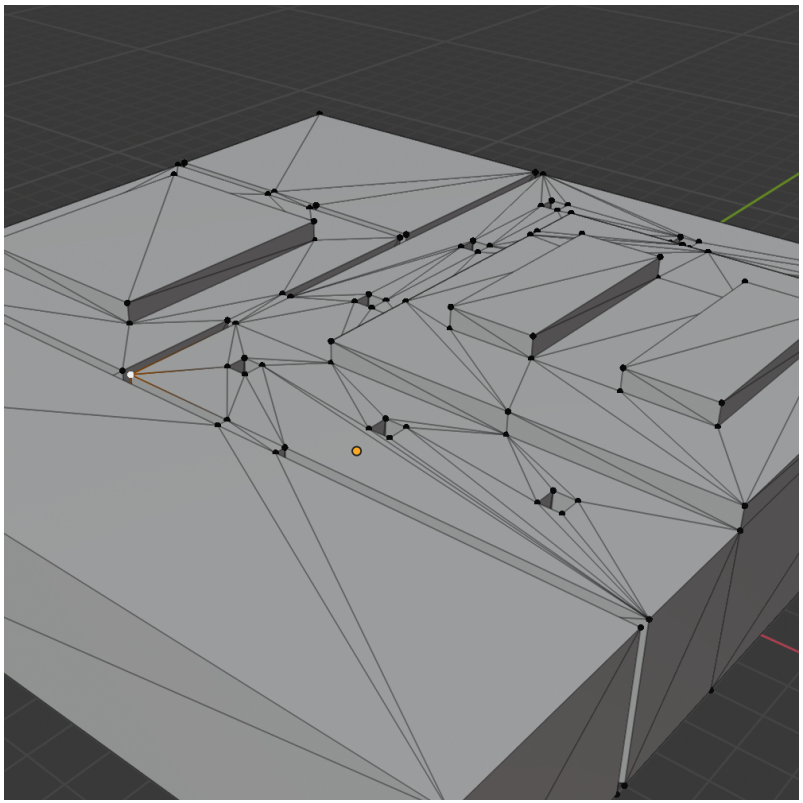
Numerical results (w/ cell marching)

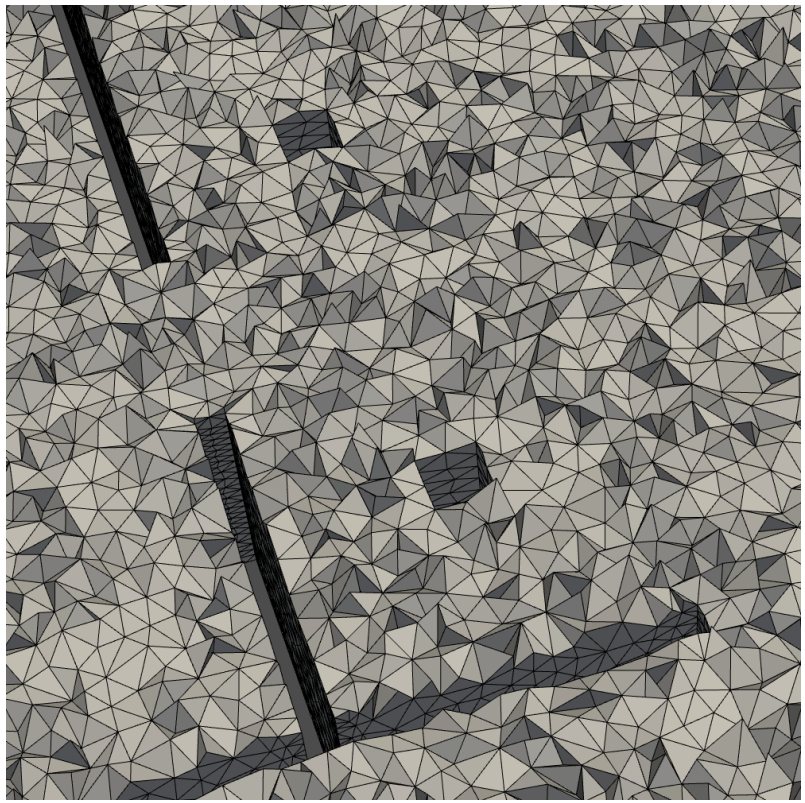


	$\tau - T$	$\tau_x - T_x$	$\tau_y - T_y$	$\tau_{xx} - T_{xx}$	$\tau_{xy} - T_{xy}$	$\tau_{yy} - T_{yy}$
Constant	3.09	3.11	3.11	2.01	2.05	2.01
Linear #1	2.99	2.43	2.40	1.39	2.01	1.39
Linear #2	2.10	1.76	1.72	0.77	1.25	0.77
Sine	2.91	1.80	1.89	0.73	1.31	0.80
Sloth	2.03	1.76	1.75	0.75	1.33	0.76

Future work

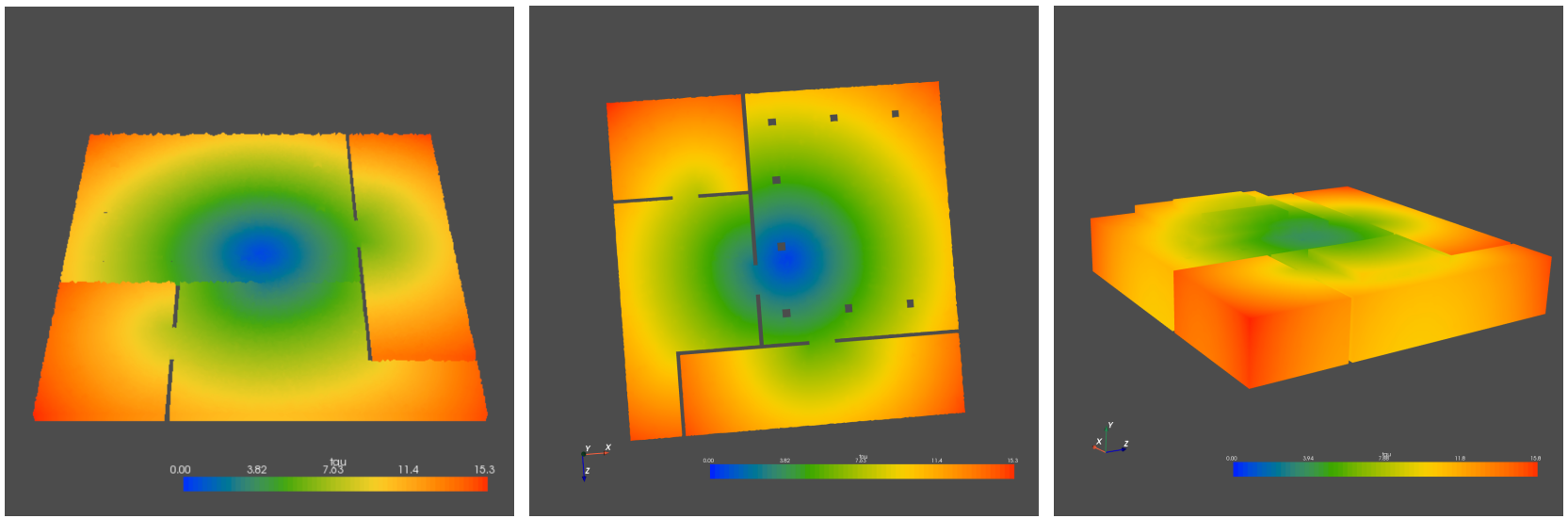
Develop JMMs for solving the eikonal equation on a tetrahedron mesh





Future work

For Hermite interpolation, use Bernstein-Bézier methods. Also need to focus on causality. Instead of marching J, may make more sense to march $\nabla\tau^2$ directly.



References

Potter, Samuel F., and Maria K. Cameron. "Jet Marching Methods for Solving the Eikonal Equation." *arXiv preprint arXiv:2009.05490* (2020).

Engquist, Björn, and Olof Runborg. "Computational high frequency wave propagation." *Acta numerica 12* (2003): 181-266.

Raghuvanshi, Nikunj, and John Snyder. "Parametric directional coding for precomputed sound propagation." *ACM Transactions on Graphics (TOG)* 37.4 (2018): 1-14.